

AI Digital Marketing Command Center

19-Agent Architecture, UI Controls & 24/7 Operations Handbook

Complete Master Architecture & 19-Agent Specification Handbook

1. Executive Summary & System Architecture

The **AI Digital Marketing Command Center** is an autonomous, multi-tenant enterprise operating system powered by **19** specialized AI Sub-Agents. It handles the entire digital marketing lifecycle for luxury chauffeur and executive transportation brands (including *Corporate Cars Melbourne* and *Opal Chauffeurs*).

High-Level System Architecture Diagram

2. Cross-Agent Interconnection & Collaborative Pipelines

The 19 AI agents do not work in isolation; they are deeply interconnected into automated growth pipelines where the output of one agent serves as the high-context input for another.

Pipeline A: The SEO & Content Flywheel

Pipeline B: Coordinated Multi-Platform Social Media Machine

3. Comprehensive Breakdown of All 19 AI Sub-Agents

Below is the complete specification of every agent, including its purpose, data source, integration points, and an exhaustive guide to every button in its interface.

1. Corporate Cars Blog Agent (`blog-agent`)

- **Category:** SEO & Content Marketing
- **Purpose:** Autonomous long-form blog writing and direct live publishing to WordPress chauffeur websites.
- **Data Source:** WordPress REST API + Claude 3.5 Sonnet + `topics.csv` queue.
- **Inter-Agent Connections:** Receives structured briefs from `seo-content-brief-agent`, triggers `internal-linking-agent` upon publishing, feeds URLs into `gsc-agent`, and reports published post counts to `monthly-report-agent`.

- Button Breakdown:

- **■ Report` Button:** Opens the live Blog Queue modal showing all 28+ planned topics, current statuses (`published`, `approved`, `drafted`), WordPress Post IDs, and direct clickable live blog URLs.
- **■ Run (Quick Trigger)` Button:** Executes the blog generation pipeline immediately for the next approved topic in queue without waiting for the scheduled morning cron.
- **■ Write (Draft Generator)` Button:** Takes the next approved topic from `topics.csv`, uses Claude to generate high-value informational copy, automatically attaches a vehicle image, and creates a WordPress draft.
- **■ Publish (Live Flip)` Button:** Checks for all drafted posts whose review window has elapsed and immediately publishes them live to WordPress with Yoast SEO metadata and Schema.org JSON-LD.
- **■ Add Blog Topic` Button:** Opens a modal to batch-import new keywords and topic ideas directly into `topics.csv` with auto-approval options.

2. Corporate Cars Social Media Agent (`corporate-cars-social-agent`)

- **Category:** Social Media Automation
- **Purpose:** Multi-platform social content generation and scheduled distribution across Instagram, Facebook, and LinkedIn.
- **Data Source:** Meta Graph API + LinkedIn API + SQLite Database (`social\_agent.db`) + Local Image Library.
- **Key Innovation:** ****Strict Coordinated Round-Robin Photo Rotation**.** Batch #1 picks Photo #1 and publishes it across Instagram, Facebook, and LinkedIn simultaneously; Batch #2 picks Photo #2 across all 3 platforms; when all photos are used, it loops back to Photo #1.
- **Button Breakdown:**
 - **■ Report` Button:** Opens the Social Media Schedule modal displaying upcoming queued posts, scheduled release times, target platform badges, and attached photo filenames.
 - **■ Run / Publish-Due` Button:** Triggers an immediate sweep of `social_agent.db` to publish any posts whose scheduled time has arrived.
 - **■ Batch Schedule Posts` Button:** Opens a campaign generator where you can paste 10+ keywords, select target platforms (IG/FB/LinkedIn), and automatically generate 30+ scheduled posts with rotating images.

3. SEO Keyword Research Agent (`seo-keyword-agent`)

- **Category:** SEO & Strategy
- **Purpose:** Discovers high-intent commercial keywords, search volume trends, keyword difficulty (KD%), and search intent (Commercial vs Informational) tailored to Melbourne suburbs and executive transport.
- **Data Source:** Search Engine Intelligence Database + Competitor Keyword Gap Analysis.
- **Inter-Agent Connections:** Feeds high-volume seed keywords directly into `seo-content-brief-agent` and `blog-agent`.
- **Button Breakdown:**
 - **■ Report` Button:** Displays discovered keyword clusters grouped by category (Airport Transfers, Corporate Chauffeur, Winery Tours, Event Hire), estimated search volume, KD difficulty score, and SERP competition ratings.
 - **■ Run / Research Keywords` Button:** Triggers AI deep-mining on a seed keyword to produce 25+ long-tail keyword variations with suburb modifiers.

4. Competitor Analysis Agent (`competitor-analysis-agent`)

- **Category:** Market Intelligence
- **Purpose:** Monitors top competing chauffeur websites across Melbourne and Sydney, comparing domain authority, estimated monthly organic traffic, content output rate, and backlink strength.
- **Data Source:** Competitive Intelligence Crawlers + SERP Scrapers.
- **Inter-Agent Connections:** Highlights competitor keyword gaps for `seo-keyword-agent` and identifies ad strategies for `competitor-ad-spy-agent`.

- Button Breakdown:

- **Report Button:** Opens the Competitor Comparison Matrix showing competitor URLs, estimated traffic share, top ranking keywords, and SWOT analysis.
- **Run Analysis Button:** Crawls specified competitor domains and recalculates market share and content gaps.

5. SEO Content Brief & Optimization Agent (`seo-content-brief-agent`)

- **Category:** Content Strategy & On-Page SEO
- **Purpose:** Creates comprehensive SEO content briefs containing target H2/H3 heading structures, semantic LSI keywords, search intent requirements, and validates Schema.org JSON-LD structured data.
- **Data Source:** SERP Analysis + Schema.org Specification + Claude LLM.
- **Inter-Agent Connections:** Acts as the editorial quality gate for `blog-agent`. Every draft generated by the Blog Agent is pre-structured and post-optimized by this agent.
- **Button Breakdown:**
 - **Report Button:** Displays the latest generated SEO content briefs, recommended word counts, required entity mentions, and Schema.org FAQ markup blocks.
 - **Generate Brief Button:** Generates an immediate content blueprint for any custom keyword input.

6. Internal Linking Agent (`internal-linking-agent`)

- **Category:** Technical & On-Page SEO
- **Purpose:** Crawls the WordPress website's pages and blog articles, detects orphaned pages, evaluates link equity distribution, and identifies contextual anchor text opportunities to link blog posts to commercial service pages.
- **Data Source:** Live WordPress REST API + `internal\_links.csv`.
- **Inter-Agent Connections:** Connects newly published articles from `blog-agent` to high-converting suburb landing pages.
- **Button Breakdown:**
 - **Report Button:** Opens the Internal Link Audit Dashboard, showing total internal links, orphaned pages with zero inbound links, and high-priority link placement recommendations.
 - **Scan & Auto-Link Button:** Executes an automated crawl of all live blog posts and suggests exact paragraph locations and anchor text to insert internal links.

7. SEO Audit Agent (`seo-audit-agent`)

- **Category:** Technical SEO
- **Purpose:** Performs 360-degree site audits checking for broken links (404s), missing meta descriptions, duplicate title tags, missing image ALT tags, slow server response times, and mobile usability issues.
- **Data Source:** Asynchronous HTTP Website Crawler + DOM Parser.
- **Inter-Agent Connections:** Feeds page health issues into `page-optimizer-agent` for instant remediation.
- **Button Breakdown:**
 - **Report Button:** Displays visual health score gauges (e.g., 94/100), critical issue alerts, warning counts, and a categorized audit table of audited URLs.
 - **Run Deep Audit Button:** Launches an asynchronous multi-threaded crawl across all website pages to detect technical SEO errors.

8. Google Search Console Agent (`gsc-agent`)

- **Category:** Real Search Analytics

- **Purpose:** Connects directly to the official **Google Search Console API** using a Google Cloud Service Account (`gsc-agent@root-clover-505611-p4.iam.gserviceaccount.com`) to extract 100% genuine, live organic search performance metrics.
- **Data Source:** **100% LIVE GOOGLE SEARCH CONSOLE API**.
- **Inter-Agent Connections:** Supplies verified organic click and impression data to `monthly-report-agent`.
- **Button Breakdown:**
 - **Report Button:** Displays live KPI cards (Total Organic Clicks, Total Impressions, Average Click-Through Rate, Average SERP Position) and a Top Performing Search Queries table with exact click counts and impressions.
 - **Refresh GSC Data Button:** Forces an immediate live API call to Google's servers to fetch the latest search performance data.

9. Google Analytics 4 Reporting Agent (`ga4-reporting-agent`)

- **Category:** Web Analytics & User Behavior
- **Purpose:** Integrates directly with the **Google Analytics Data API (v1beta)** to monitor live visitor sessions, user acquisition channels, average engagement duration, and goal conversions for Property ID `550393874`.
- **Data Source:** **100% LIVE GOOGLE ANALYTICS 4 DATA API** (Measurement ID: `G-2CM2BW6QKN`).
- **Inter-Agent Connections:** Feeds genuine user session and landing page conversion data into `monthly-report-agent`.
- **Button Breakdown:**
 - **Report Button:** Opens the custom GA4 Live Overview modal with 4 executive KPI cards (Total Users, Total Sessions, Conversions, Engagement Rate), Acquisition Channel Breakdown bars, Top Landing Pages table, and the Live Stream Status indicator.
 - **Query Live GA4 API Button:** Executes a direct `runReport` API request to Google Analytics Data API to retrieve fresh analytics.

10. Google Ads Monitoring Agent (`google-ads-monitoring-agent`)

- **Category:** Paid Search (PPC)
- **Purpose:** Monitors Google Search ad campaigns, keyword Quality Scores, Cost-Per-Click (CPC), Search Impression Share, and identifies wasted spend on negative search terms.
- **Safety Feature:** **Protected Live Spend Guard** (`ADS_LIVE_EXECUTION_ENABLED=false`) ensures zero financial risk while delivering real-time strategy recommendations.
- **Button Breakdown:**
 - **Report Button:** Displays Google Ads campaign status, budget utilization, recommended negative keywords, and Quality Score optimizations.
 - **Run Monitoring Sweep Button:** Audits active ad groups and generates high-converting bid adjustment recommendations.

11. Google Ads Optimization Agent (`google-ads-optimization-agent`)

- **Category:** Paid Search Optimization
- **Purpose:** Generates high-CTR Responsive Search Ads (RSAs), winning headline combinations (e.g., "Melbourne Chauffeur From \$95 | Fixed Airport Pricing"), Callout Extensions, and Structured Snippets.
- **Inter-Agent Connections:** Uses ad intelligence from `competitor-ad-spy-agent` to create market-beating ad copy.
- **Button Breakdown:**
 - **Report Button:** Opens the RSA Copy Matrix showing 15 headlines and 4 descriptions sorted by estimated ad strength.
 - **Optimize Ad Copy Button:** Generates fresh, high-relevance ad variations targeted to executive travelers and airport commuters.

12. Meta Ads Monitoring Agent (`meta-ads-monitoring-agent`)

- **Category:** Social Advertising (PPC)
- **Purpose:** Monitors Facebook & Instagram Ads campaigns, Return on Ad Spend (ROAS), Cost Per Lead (CPL), audience frequency, and creative ad fatigue.
- **Button Breakdown:**
 - **Report Button:** Displays Meta campaign performance cards, audience demographic breakdown, and ad creative fatigue alerts.
 - **Audit Ad Spend Button:** Analyzes audience saturation and recommends fresh creative swaps.

13. Social Media Analytics Agent (`social-analytics-agent`)

- **Category:** Social Media Performance
- **Purpose:** Queries Meta Graph API and LinkedIn Marketing API to track organic post engagement, follower growth, reach, top-performing post formats, and audience engagement spikes.
- **Inter-Agent Connections:** Feeds published post engagement numbers into `monthly-report-agent`.
- **Button Breakdown:**
 - **Report Button:** Displays multi-platform organic performance charts, total likes, shares, comments, and top-performing post rankings.
 - **Sync Social Metrics Button:** Fetches the latest engagement statistics directly from Instagram, Facebook, and LinkedIn.

14. Review & Reputation Management Agent (`reputation-agent`)

- **Category:** Brand Reputation & Local SEO
- **Purpose:** Monitors Google Business Profile reviews, TripAdvisor feedback, and Facebook ratings. Calculates overall sentiment score (e.g., 4.9/5.0) and generates professional, polite AI replies to customer reviews.
- **Button Breakdown:**
 - **Report Button:** Displays average star rating, total review count, sentiment distribution (Positive / Neutral / Negative), and recent customer reviews with AI response drafts.
 - **Scan New Reviews Button:** Checks review platforms for new ratings and drafts suggested reply templates for admin approval.

15. Lead Management Agent (`lead-management-agent`)

- **Category:** CRM & Revenue Pipeline
- **Purpose:** Ingests and scores inbound quotation requests and website inquiries. Calculates lead priority (High / Medium / Low), categorizes booking types (Airport Transfer, Corporate VIP, Wedding, Winery Tour), and tracks total pipeline revenue value.
- **Inter-Agent Connections:** Aggregates total monthly inbound leads and pipeline value into `monthly-report-agent`.
- **Button Breakdown:**
 - **Report Button:** Opens the custom visual Lead Management Dashboard with 4 KPI cards (Total Pipeline Value, Active Inbound Leads, Corporate Accounts, Average Lead Score), Priority Leads table with contact details and booking values, and Actionable Sales Checklists.
 - **Score New Leads Button:** Runs AI evaluation on newly received inquiries to assign lead priority scores and identify VIP corporate executives.

16. Monthly Marketing Report Agent (`monthly-report-agent`)

- **Category:** Executive Analytics & Consolidation
- **Purpose:** The master aggregator of the entire system. Dynamically collects live data from ALL 18 other agents (real blog posts from WordPress, social posts from DB, organic clicks from GSC API, user traffic from GA4, SEO audits, and lead value) to generate consolidated executive performance summaries.
- **Button Breakdown:**
 - **Report Button:** Opens the custom Executive Marketing Dashboard showing 4 top KPI cards (Consolidated Organic Clicks, Live GA4 Sessions, Content Produced, Estimated Pipeline Value), a 5-Channel Performance Breakdown table, and Participating Verified Agents badge strip.
 - **Instant MTD Report (Today) Button:** Generates an immediate Month-to-Date executive report with exact figures up to the current day.
 - **Full Month Cycle Sync Button:** Runs a full 30-day cross-agent consolidation cycle.

17. External Link Building Agent (`external-link-building-agent`)

- **Category:** Off-Page SEO & Backlinks
- **Purpose:** Identifies high-authority niche travel, luxury lifestyle, and Melbourne corporate directories for white-hat backlink acquisition. Drafts personalized outreach emails and tracks link placement status.
- **Button Breakdown:**
 - **Report Button:** Displays target outreach opportunities, Domain Authority (DA) scores, outreach pipeline statuses (Discovered, Outreached, Placed), and live backlink counts.
 - **Run Daily Outreach Batch Button:** Generates 5 personalized outreach email templates targeted to high-DA website editors.

18. Competitor Ad Spy Agent (`competitor-ad-spy-agent`)

- **Category:** Competitive Intelligence
- **Purpose:** Reverse-engineers live and historical Google Search and Meta ads run by competitor chauffeur services. Extracts winning headlines, pricing hooks, primary keywords, and visual ad formats.
- **Button Breakdown:**
 - **Report Button:** Opens the Ad Spy Dashboard showing active competitor ads, estimated monthly ad spend, top targeted keywords, and recommended winning ad copy templates.
 - **Spy Ads Button:** Initiates deep intelligence extraction on specified competitor websites.

19. Page SEO Doctor & Algorithm Optimizer Agent (`page-optimizer-agent`)

- **Category:** Core Web Vitals & On-Page Optimization
- **Purpose:** Performs DOM-level page health diagnostics, verifies keyword density, validates OpenGraph social preview tags, checks mobile responsive viewport tags, and optimizes meta tags to align with Google's latest Helpful Content & Core Algorithm updates.
- **Button Breakdown:**
 - **Report Button:** Displays per-page optimization scores (e.g., 96/100), title tag length checks, meta description sentiment, and Schema markup validation.
 - **Optimize Page Button:** Analyzes a target landing page and outputs exact code recommendations to fix on-page SEO issues.

4. Command Center UI, Navigation & Global Controls

1. Top Navigation Bar Features

- **Multi-Website Dropdown** (`#websiteSelector`):

Allows instant 1-click switching between different chauffeur brands (e.g., **Corporate Cars Melbourne** vs **Opal Chauffeurs**). All agent telemetry, topics, databases, and metrics automatically switch context.

- **System Health Badge:**

Real-time indicator showing engine status, database connection, and active agent count (**19 Agents Active**).

- **Auto-Refresh Toggle:**

Automatically refreshes active metrics every 30 seconds without full-page reloads.

- **Admin Profile & PIN Protection:**

Secures the system using role-based access control. Sensitive actions (adding campaigns, executing live publishing, modifying API keys) require admin verification (Default PIN: `26032024`, Admin: `sonutripathi9305@gmail.com`).

2. Main Sidebar Views

- **Cockpit / Overview:** Displays high-level system KPIs, AI token consumption gauges, live activity feed, and multi-channel performance charts.

- **Sub-Agents Grid:** Displays all 19 AI agents as interactive glassmorphism cards with category badges, status indicators, and one-click action buttons.

- **Task Queue** (`/api/tasks`): Real-time monitoring of all queued, running, completed, and failed tasks with exact start times and execution duration.

- **Error Tracking** (`/api/errors`): Zero-error diagnostic center providing automated retry buttons and failure tracebacks.

- **Structured Logs** (`/api/logs`): Filterable log stream allowing inspection of central system logs or isolated individual agent logs.

- **Security Audit Trail** (`/api/audit-trail`): Immutable chronological record of every administrative action, task creation, and system configuration change.

- **System Settings:** Controls AI Model Router assignments (Claude 3.5 Sonnet, DeepSeek, OpenAI GPT-4o), Ads Safety Guard (`ADS_LIVE_EXECUTION_ENABLED`), and Live Publishing permissions.

5. Cloud Infrastructure & 24/7 Autonomous Operation

1. Production Hosting on Render.com

- **Platform:** **Render.com Cloud Web Service**

- **Live Production URL:** `https://ai-digital-marketing-gm68.onrender.com`

- **GitHub Continuous Deployment:** Connected to `https://github.com/sonutripathi123/ai-digital-marketing.git` (`main` branch). Any pushed update is automatically built and deployed to the cloud within 60 seconds.

2. Autonomous Background Schedulers & Keep-Alive Engines

- **FastAPI Background Runner Daemon** (`SchedulerManager`):

Runs inside the production Python process, evaluating cron expressions every 15 seconds.

- **Self-Healing Watchdog Engine** (`check_and_auto_catchup_daily_blog`):

Runs on startup and every 15 minutes. If a server reboot or network lag caused the 10:00 AM IST scheduled blog run to be skipped, the watchdog automatically detects that today's blog is missing and immediately generates and publishes it live to WordPress without waiting for the next day.

- Social Media 5-Minute Daemon (`publish-due`):

Sweeps `social\_agent.db` every 5 minutes and publishes due posts across Meta (Facebook & Instagram) and LinkedIn according to the strict round-robin photo rotation.

- 24/7 Keep-Alive Ping (`cron-job.org`):

Pings `https://ai-digital-marketing-gm68.onrender.com/health` every 10 minutes to prevent cloud instance sleeping, ensuring 100% uninterrupted 24/7/365 autonomous operation.

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